

EDWARD VARVAK

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EDUCATION

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| University of Chicago
<i>Doctor of Philosophy (Ph.D.) in Mathematics</i> , advised by Ben Bakker | <i>Sept 2026 - May 2027</i> |
| University of Illinois Chicago
<i>Doctor of Philosophy (Ph.D.) in Mathematics</i> , advised by Ben Bakker, | <i>Sept 2022 - Sept 2026 (transfer)</i> |
| Stony Brook University
<i>Bachelor of Science (B.S.) in Mathematics</i>
<i>Bachelor of Science (B.S.) in Applied Mathematics and Statistics</i> | <i>Sept 2018 - May 2022</i> |

RESEARCH INTERESTS

Complex algebraic geometry, Hodge theory, Abelian varieties, symplectic varieties, Lagrangian fibrations, algebraic foliations, monodromy representations, non-Abelian Hodge theory.

RESEARCH AND PUBLICATIONS

1. *The Mumford–Tate group of a very general Lagrangian fiber*, 2026
2. *The Monodromy of Compact Lagrangian Fibrations*, 16 pages, **IMRN**, rnag057, 2026
3. *Recovering Plane Nodal Curves from their Bitangents*, 12 pages, **Undergraduate Honors Thesis**, advised by Olivier Martin, 2022.

INVITED TALKS

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| <i>GTA Philadelphia</i> , Temple University | <i>May 2026</i> |
| <i>Research Seminar</i> , Institute for Advanced Study | <i>November 2025</i> |
| <i>Graduate Student Colloquium</i> , University of Illinois Chicago | <i>February 2025</i> |

ORGANIZATION AND SERVICE

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| Co-organized Graduate Algebraic Geometry Seminar (GAGS) | <i>Fall 2025-Spring 2026</i> |
| Judged for QED Competition (Math Circles of Chicago) | <i>December 2025</i> |
| Organized Weil Conjecture Reading Group | <i>Fall 2025</i> |
| Organized Lie Groups Seminar | <i>Fall 2024</i> |

HONORS AND AWARDS

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| Bernard Kurtin Award for research in mathematics, UIC MSCS Department | <i>2025-2026</i> |
| UIC RTG Grant | <i>Fall 2025,</i>
<i>Summer 2025,</i>
<i>Spring-Summer 2024,</i>
<i>Spring-Summer 2023</i> |
| Honors in Mathematics, SBU Mathematics Department | <i>2018-2022</i> |
| Kuga-Sah Award for excellence in mathematics, SBU Mathematics Department | <i>2022</i> |
| Cum Laude, Stony Brook University | <i>2022</i> |

TEACHING EXPERIENCE

University of Chicago:

Math 15200: Calculus II (Graduate Student Lecturer)

Fall 2026

University of Illinois Chicago:

Math 181: Calculus II (Teaching Assistant)

Spring 2026

Math 160: Finite Math for Business (Teaching Assistant)

Spring 2025,

Fall 2024,

Fall 2023,

Fall 2022

LANGUAGES

English (Native)

Russian (Fluent)

REFERENCES

Ben Bakker (Doctoral Advisor)

Professor

University of Chicago

bakker.uchicago@gmail.com

Valentino Tosatti

Professor

Courant Institute

tosatti@cims.nyu.edu

İzzet Coşkun

LAS Distinguished Professor

University of Illinois Chicago

icoskun@uic.edu

McKinley Meyer (teaching reference)

Lecturer

University of Illinois Chicago

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